

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 4 -- Ensuring Successful Operation of a Cloud Solution -- Visual Maps

GCP Study Notes

Study Notes April 2026

1. Snapshot vs Image Comparison Visual

SNAPSHOT vs IMAGE -- SIDE BY SIDE

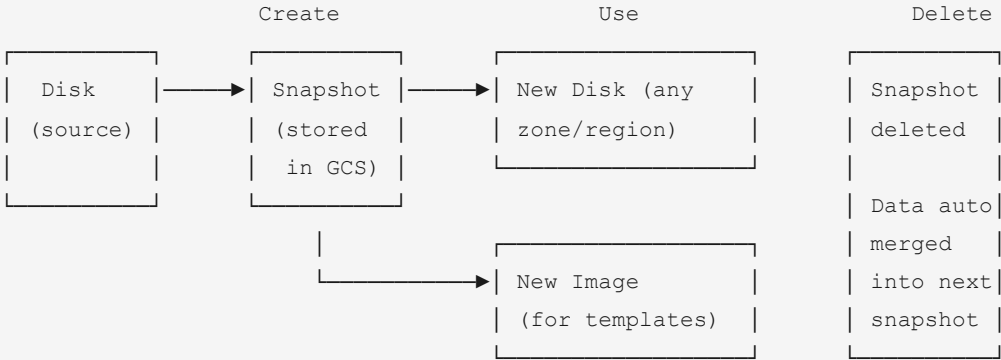
SNAPSHOT	IMAGE
Purpose: BACKUP & DISASTER RECOVERY	Purpose: CREATE NEW VMs / BOOT DISKS
Boot Disk (YES) Data Disk (YES)	Boot Disk ONLY
Works on ANY persistent disk	Only from boot disks
Storage: INCREMENTAL	Storage: FULL COPY
Snap1 = Full copy Snap2 = Only changes since Snap1 Snap3 = Only changes since Snap2	Complete image every time Image Families: YES my-app-family - my-app-v1 - my-app-v2 - my-app-v3 (latest)
Families: NO	
Schedule: YES (automated daily/weekly/hourly)	Schedule: NO (manual creation)
Create VM from: SLOWER (restore to disk, then boot)	Create VM from: FASTER (directly boot from image)
Cross-region: YES (restore disk in any zone)	Cross-region: YES (create VM in any zone)

WHEN TO USE WHICH?

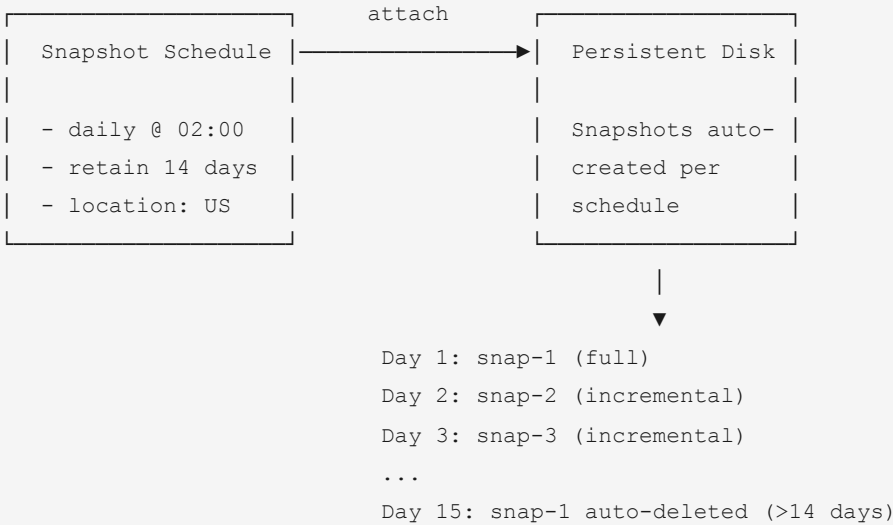
"I need to back up my database disk before an upgrade"
→ SNAPSHOT

- "I need daily automated backups with 14-day retention"
→ SNAPSHOT SCHEDULE
- "I need a golden image with my app pre-installed for MIG templates"
→ CUSTOM IMAGE (with image family)
- "I need to clone a disk to a different region for DR"
→ SNAPSHOT (create disk from snapshot in target region)
- "I need to create 100 identical VMs quickly"
→ IMAGE (in instance template)

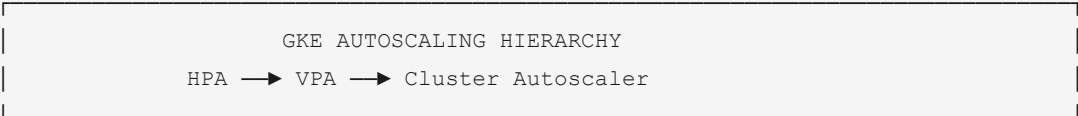
SNAPSHOT LIFECYCLE



SNAPSHOT SCHEDULE SETUP

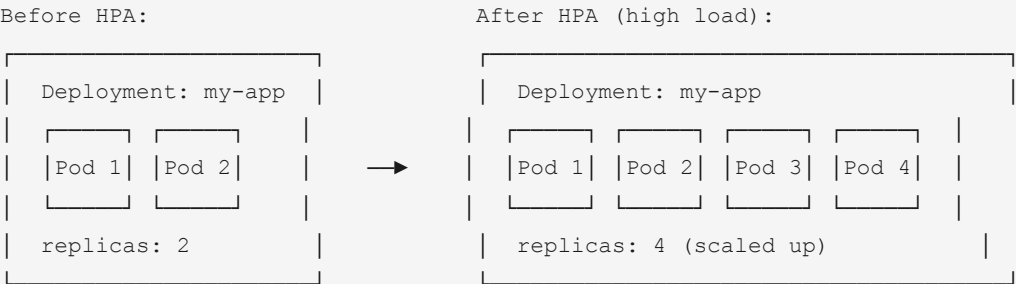


2. GKE Autoscaling Hierarchy



LEVEL 1: HPA (Horizontal Pod Autoscaler)

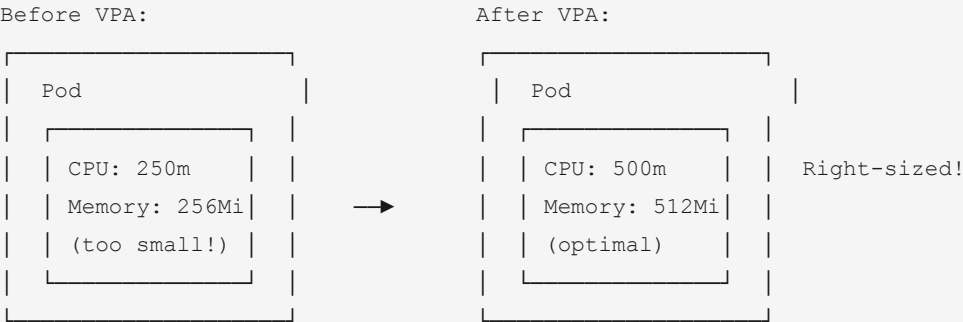
Scales: NUMBER OF POD REPLICAS
Based on: CPU, memory, custom metrics



Config: min=2, max=10, target CPU=70%

LEVEL 2: VPA (Vertical Pod Autoscaler)

Scales: CPU/MEMORY REQUESTS PER POD
Based on: Historical resource usage

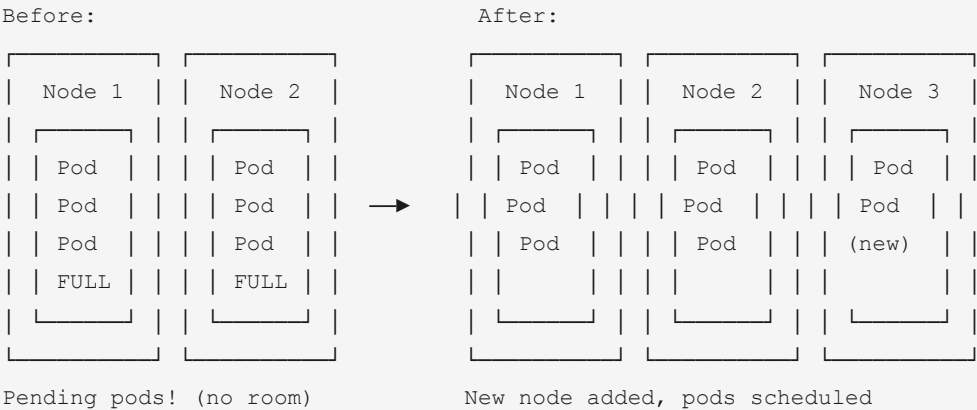


VPA Modes:

Off	Recommendations only (no changes)
Initial	Set resources only at pod creation
Recreate	Evict + recreate pods with new resources
Auto	Like Recreate (in-place when available)

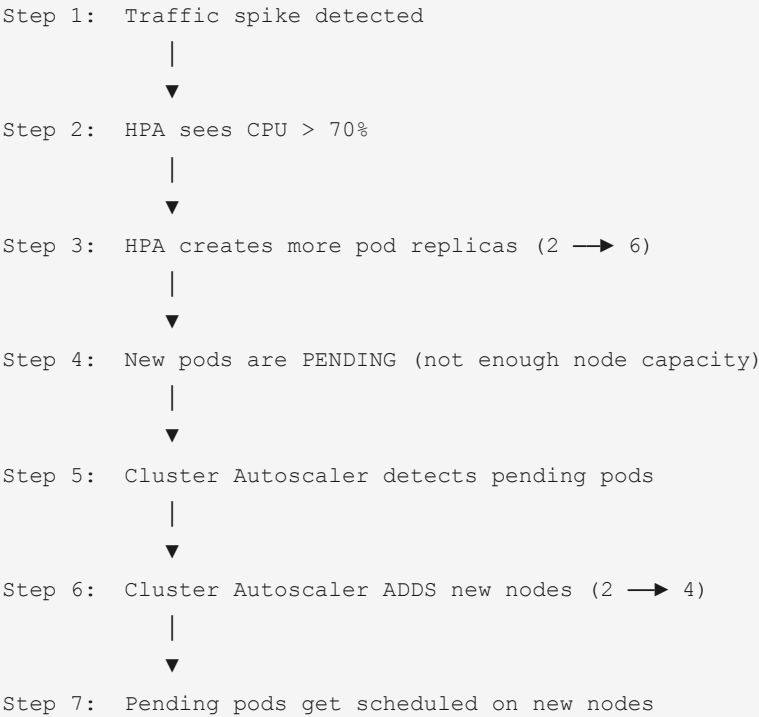
LEVEL 3: CLUSTER AUTOSCALER

Scales: NUMBER OF NODES in a node pool
Based on: Pending (unschedulable) pods



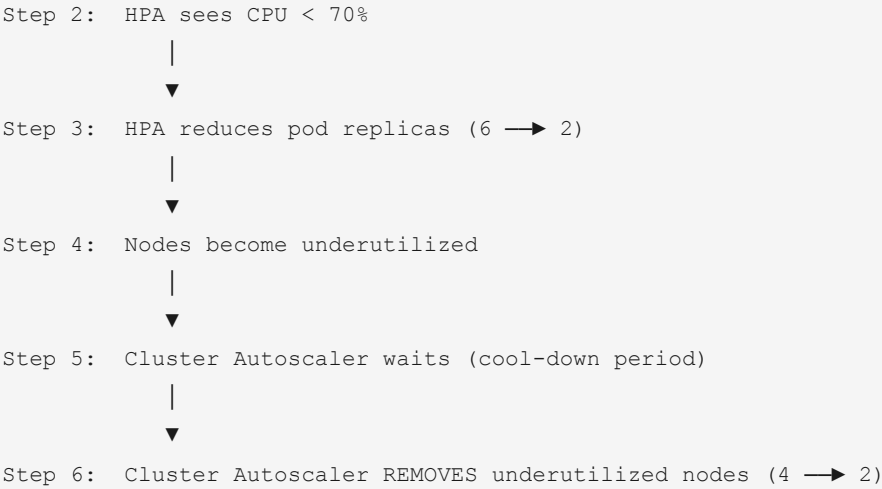
HOW ALL THREE WORK TOGETHER

Traffic INCREASES:



Traffic DECREASES:





AUTOSCALER COMPARISON		
Autoscaler	What It Scales	Trigger Signal
HPA	Pod replica count	CPU, memory, custom metrics
VPA	Pod CPU/mem request	Historical resource usage
Cluster AS	Node count	Pending (unschedulable) pods

NOTE: Do NOT use HPA and VPA on the same metric (e.g., both on CPU).
They will conflict. Use HPA for scaling out, VPA for right-sizing.

3. Cloud Run Traffic Splitting Visual



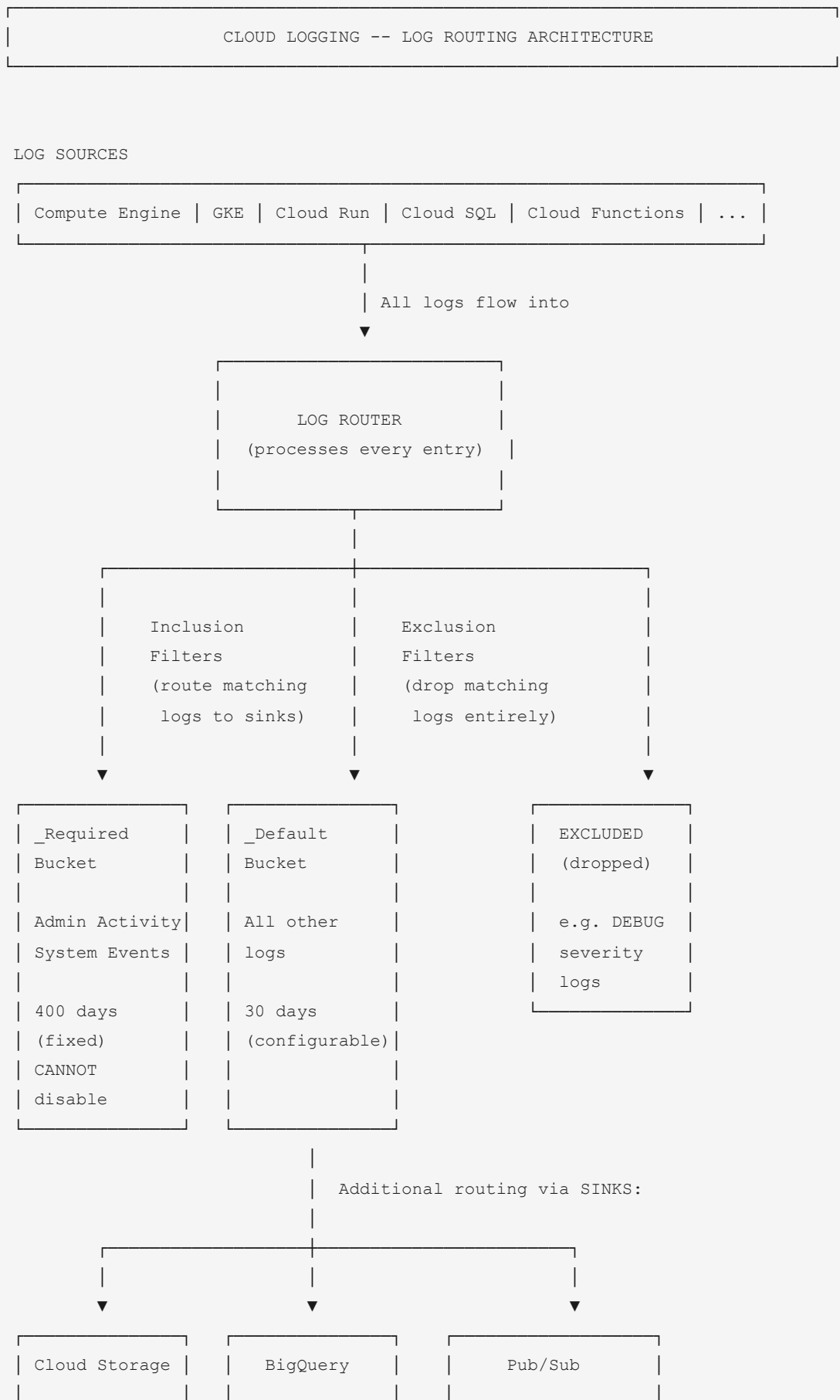
Commands:

```
| # Step 1: Deploy without traffic |  
| gcloud run deploy my-service --image=v3 --no-traffic |  
| |  
| # Step 2: Send 5% to canary |  
| gcloud run services update-traffic --to-revisions=v2=95,v3=5 |  
| |  
| # Step 3: Monitor metrics, increase gradually |  
| gcloud run services update-traffic --to-revisions=v2=50,v3=50 |  
| |  
| # Step 4: Complete rollout |  
| gcloud run services update-traffic --to-revisions=v3=100 |  
| |  
| # ROLLBACK (if issues detected at any step): |  
| gcloud run services update-traffic --to-revisions=v2=100 |
```

TAGGED REVISIONS (testing without affecting production traffic):

```
| |  
| Main URL: my-service-xxx.a.run.app → v2 (100% production) |  
| |  
| Tag URL:  canary---my-service-xxx.a.run.app → v3 (testing only) |  
| |  
| The tag URL lets you test v3 WITHOUT any production traffic |  
| going to it. Only explicit requests to the tag URL reach v3. |
```


4. Log Routing Architecture



Long-term	SQL analysis	Real-time
archival	& dashboards	processing
Compliance	Reporting	Forward to
		SIEM / on-prem
Cheapest	Queryable	Third-party tools

|



Custom Log
Bucket
Configurable
retention
(1-3650 days)
Log Analytics
(SQL queries)

SINK SETUP PROCESS

Step 1: Create the sink

```
gcloud logging sinks create my-bq-sink \
  bigquery.googleapis.com/projects/PROJ/datasets/logs_dataset \
  --log-filter='resource.type="gce_instance"'
```

|



Step 2: Get the writer identity (service account)

```
gcloud logging sinks describe my-bq-sink \
  --format="get(writerIdentity)"
```

Output: serviceAccount:pl23-456@gcp-sa-logging.iam.gserviceaccount

|



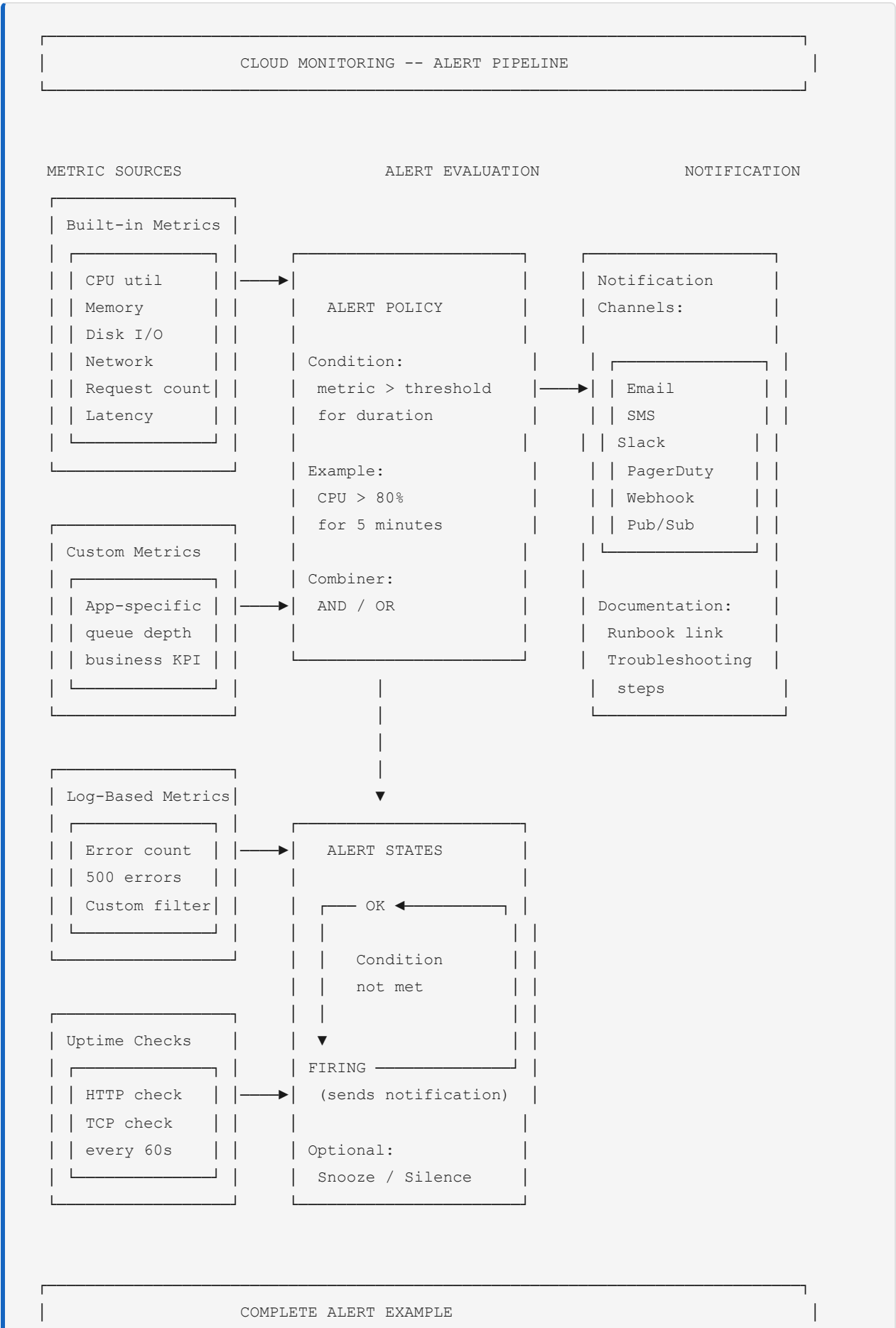
Step 3: Grant the writer identity access to the destination

```
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="WRITER_IDENTITY" \
  --role="roles/bigquery.dataEditor"
```

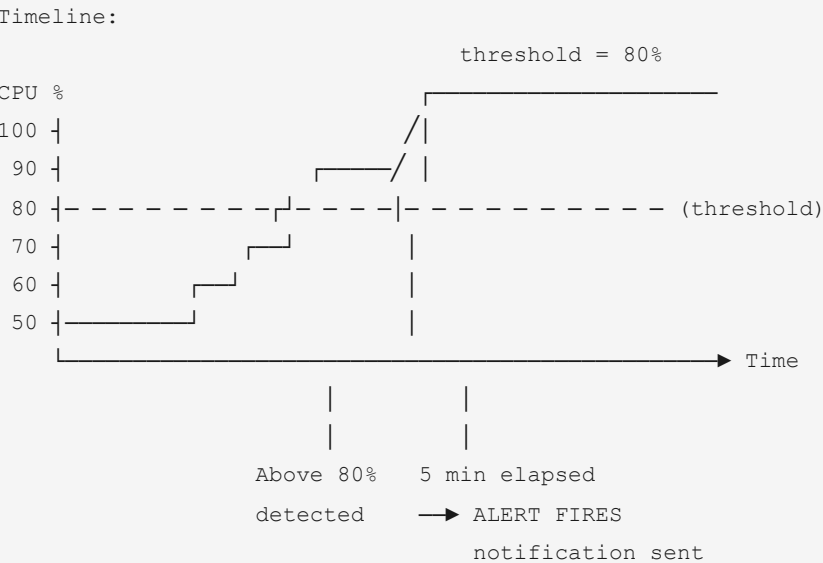
LOG BUCKET COMPARISON

Bucket	Retention	Notes
_Required	400 days (fixed)	Always on, cannot disable, free Admin Activity + System Events
_Default	30 days (configurable)	All other logs, configurable retention Can add exclusion filters
Custom	1-3650 days (configurable)	User-created, Log Analytics support Can be locked (prevent deletion)

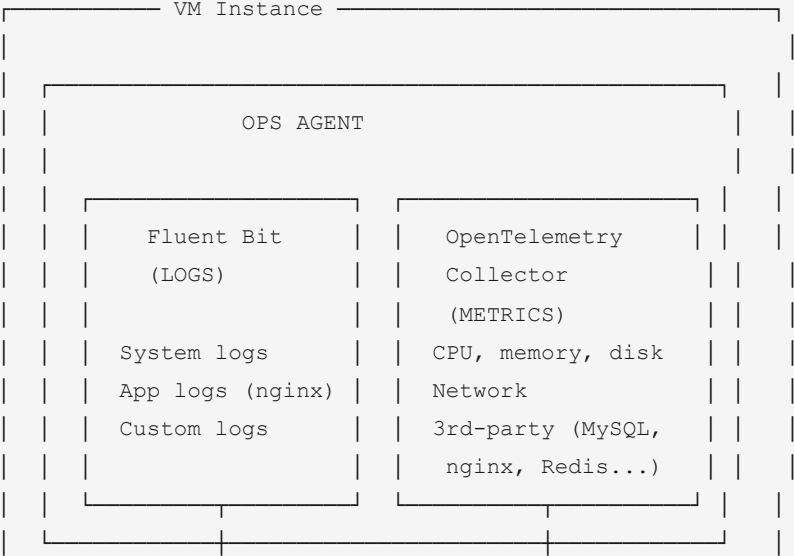
5. Monitoring Alert Pipeline Flow

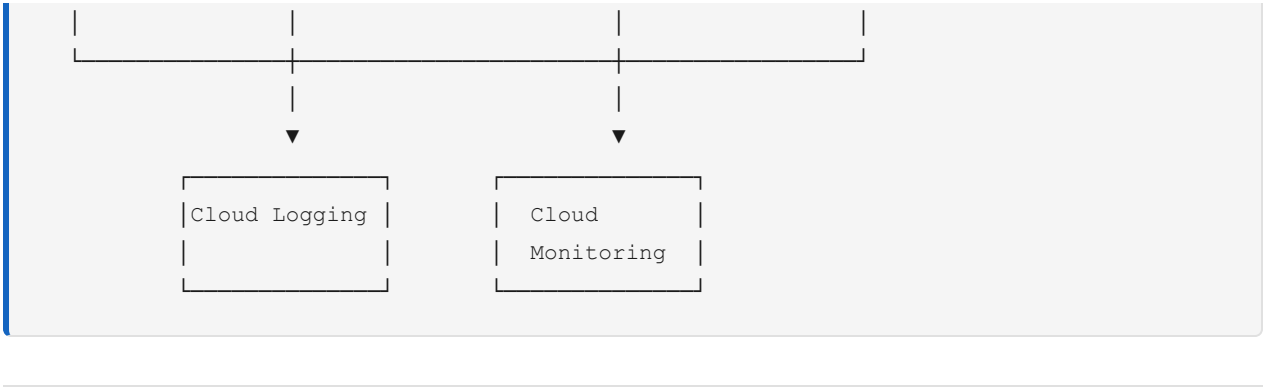


```
| Alert: "High CPU on Production VMs" |
|                                     |
| Metric:   compute.googleapis.com/instance/cpu/utilization |
| Filter:   resource.type = "gce_instance" |
| Threshold: > 0.8 (80%) |
| Duration: 300 seconds (5 minutes) |
| Alignment: 60s, ALIGN_MEAN |
|                                     |
| Notification: Email to admin@example.com + Slack #ops-alerts |
| Documentation: "CPU above 80%. Check for runaway processes." |
```



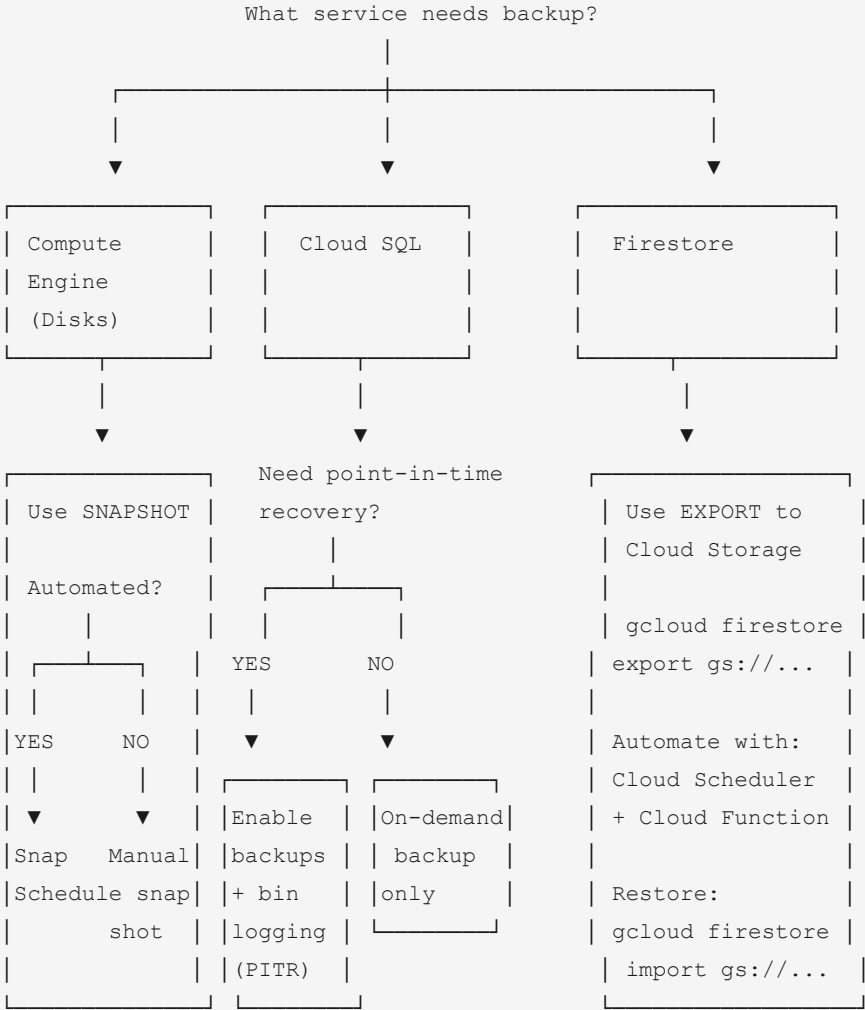
OPS AGENT ARCHITECTURE





6. Backup and Restore Decision Tree

BACKUP & RESTORE DECISION TREE



CLOUD SQL BACKUP TYPES

TYPE 1: AUTOMATED BACKUPS

Mon	Tue	Wed	Thu	Fri	Sat	Sun
02:00	02:00	02:00	02:00	02:00	02:00	02:00

--backup-start-time=02:00

--retained-backups-count=7

TYPE 2: ON-DEMAND BACKUP

=====

gcloud sql backups create --instance=my-instance

(manual, any time, for pre-upgrade safety)

TYPE 3: POINT-IN-TIME RECOVERY (PITR)

=====

Binary Log / WAL (continuous recording)

▲

Can restore to ANY SECOND within retention window

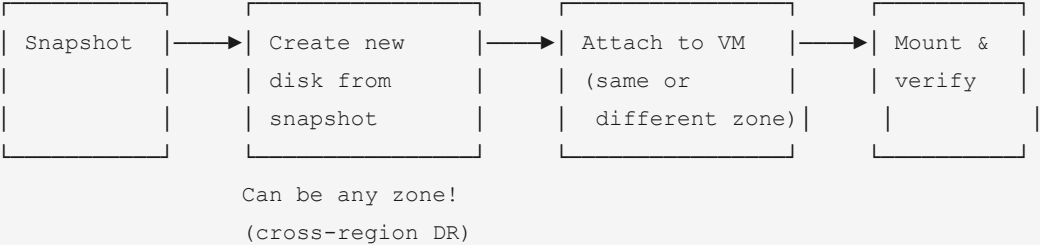
gcloud sql instances clone my-inst restored \

--point-in-time="2024-06-15T10:30:00.000Z"

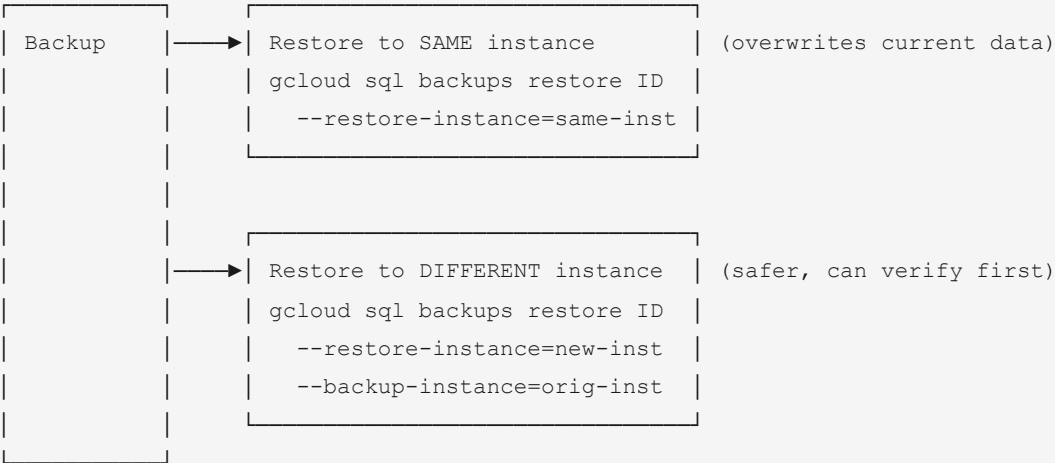
Requires: --enable-bin-log (MySQL) or WAL archiving (PostgreSQL)

RESTORE PATHS

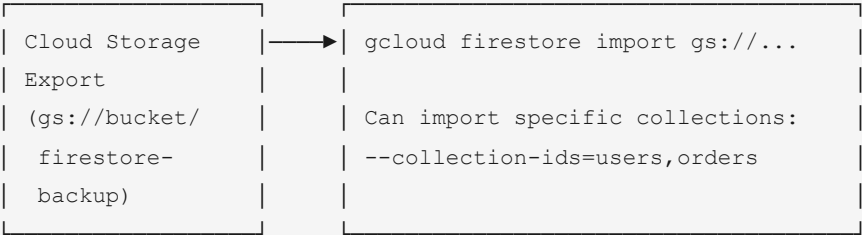
COMPUTE ENGINE DISK RESTORE:



CLOUD SQL RESTORE:



FIRESTORE RESTORE:



BACKUP STRATEGY COMPARISON			
Service	Backup Method	Automated?	Point-in-Time Recovery?
Compute Engine (disks)	Snapshots (incremental)	Yes (schedules)	No (use snapshot closest to time)
Cloud SQL	Built-in backups	Yes (daily)	Yes (binary log / WAL)
Firestore	Export to GCS	No (DIY Scheduler)	No (restore from export only)
Spanner	Built-in backups	Yes	Yes
AlloyDB	Built-in backups	Yes	Yes (continuous)

7. Audit Log Types Visual

AUDIT LOG TYPES

ADMIN ACTIVITY LOGS

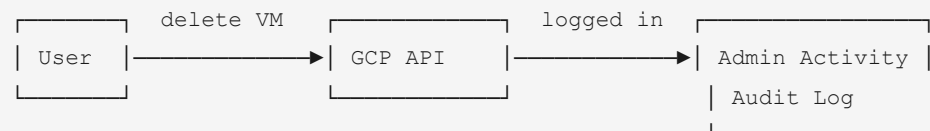
What: API calls that MODIFY resources (create, delete, update, patch)

Enabled: ALWAYS ON (cannot be disabled)

Cost: FREE

Retention: 400 days (_Required bucket)

Example: Someone deletes a VM, changes IAM policy, creates a bucket



DATA ACCESS LOGS

What: API calls that READ data or metadata

Enabled: MUST enable manually (except BigQuery: always on)

Cost: CHARGED (can generate high volume)

Retention: 30 days (_Default bucket) unless routed elsewhere

Example: Someone reads a file in Cloud Storage, queries BigQuery

Types: ADMIN_READ, DATA_READ, DATA_WRITE

SYSTEM EVENT LOGS

What: Google-initiated system actions

Enabled: ALWAYS ON (cannot be disabled)

Cost: FREE

Example: Live migration, auto-scaling, auto-repair events

POLICY DENIED LOGS

What: Security policy violations (denied API calls)

Enabled: ALWAYS ON

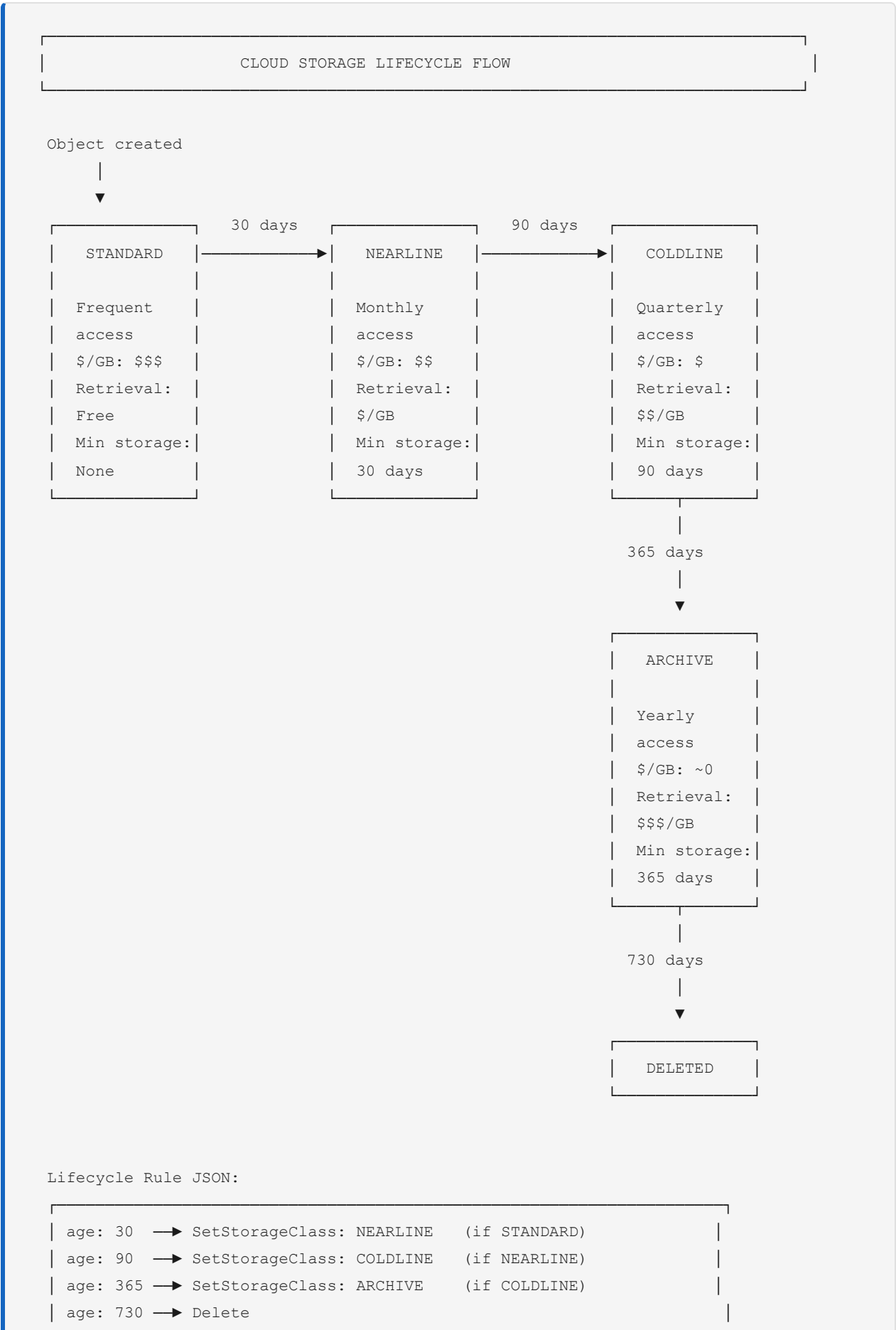
Cost: FREE

Example: User tries to access a resource without permission

SUMMARY TABLE:

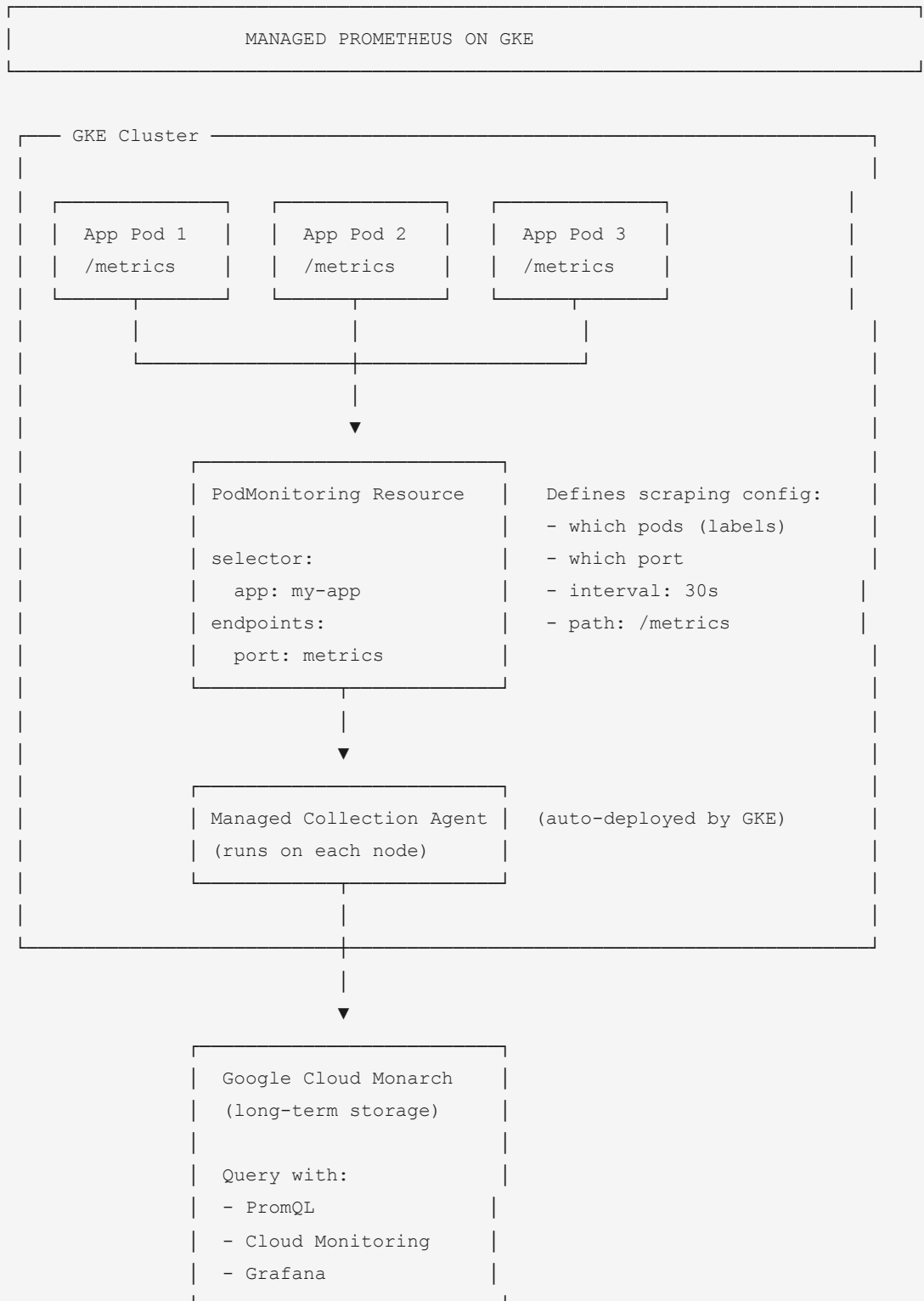
Log Type	Default State	Cost	Disable?
Admin Activity	Always ON	Free	No
Data Access	OFF (enable)	Charged	Yes
System Event	Always ON	Free	No
Policy Denied	Always ON	Free	No

8. Cloud Storage Lifecycle Management



```
isLive: false, numNewerVersions: 3 → Delete old versions
```

9. Managed Prometheus on GKE



10. Networking Operations Summary

NETWORKING OPERATIONS -- KEY TASKS

EXPANDING A SUBNET:

Before: 10.0.1.0/24 (252 usable IPs)

After: 10.0.0.0/20 (4,092 usable IPs)

Rules:

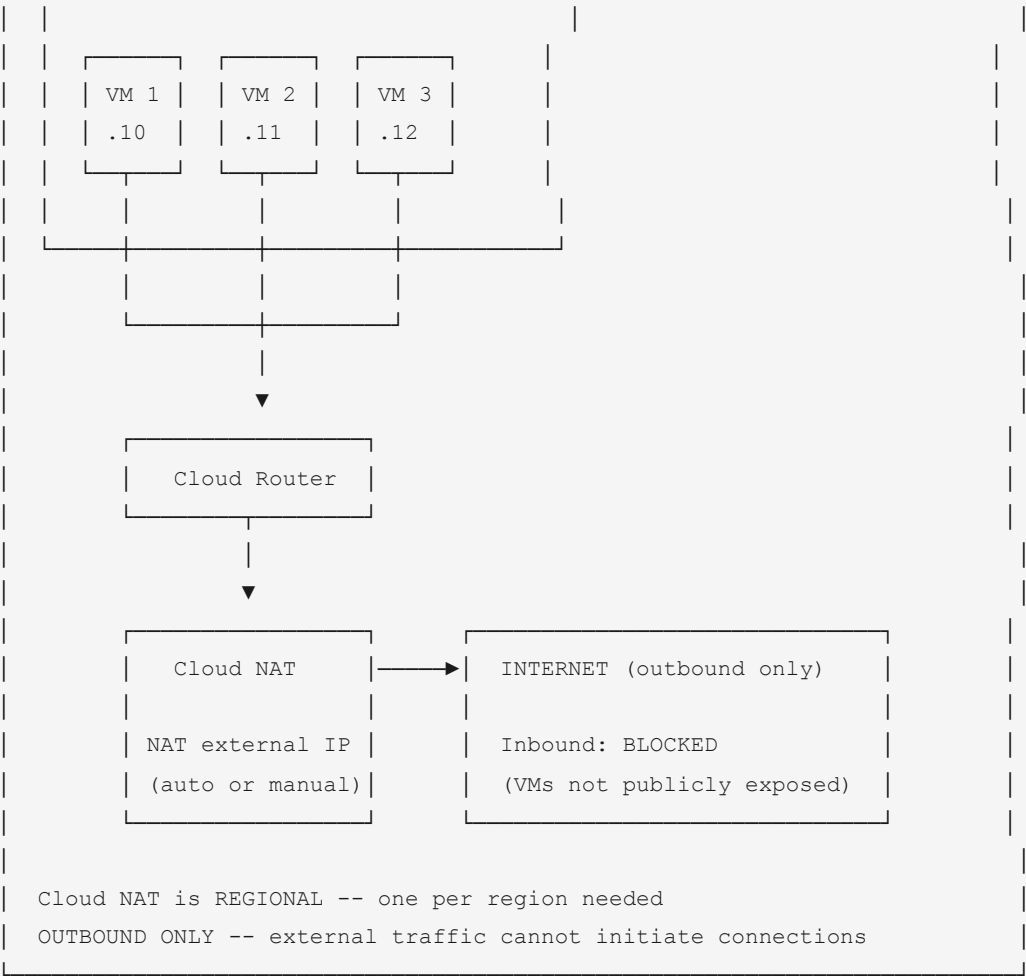
- Can only EXPAND (never shrink)
- New range must CONTAIN original range
- NON-DISRUPTIVE (no downtime)
- Cannot overlap with other subnets

STATIC IP TYPES:

EPHEMERAL	STATIC
External:	External:
Released when VM stops	Persists after VM stops/deletes
Internal:	Internal:
Auto-assigned from subnet	Specific IP reserved
Cost: Free	Cost: Charged if UNUSED
Use: Dev/test	Use: Production DNS, LB

CLOUD NAT MODEL:

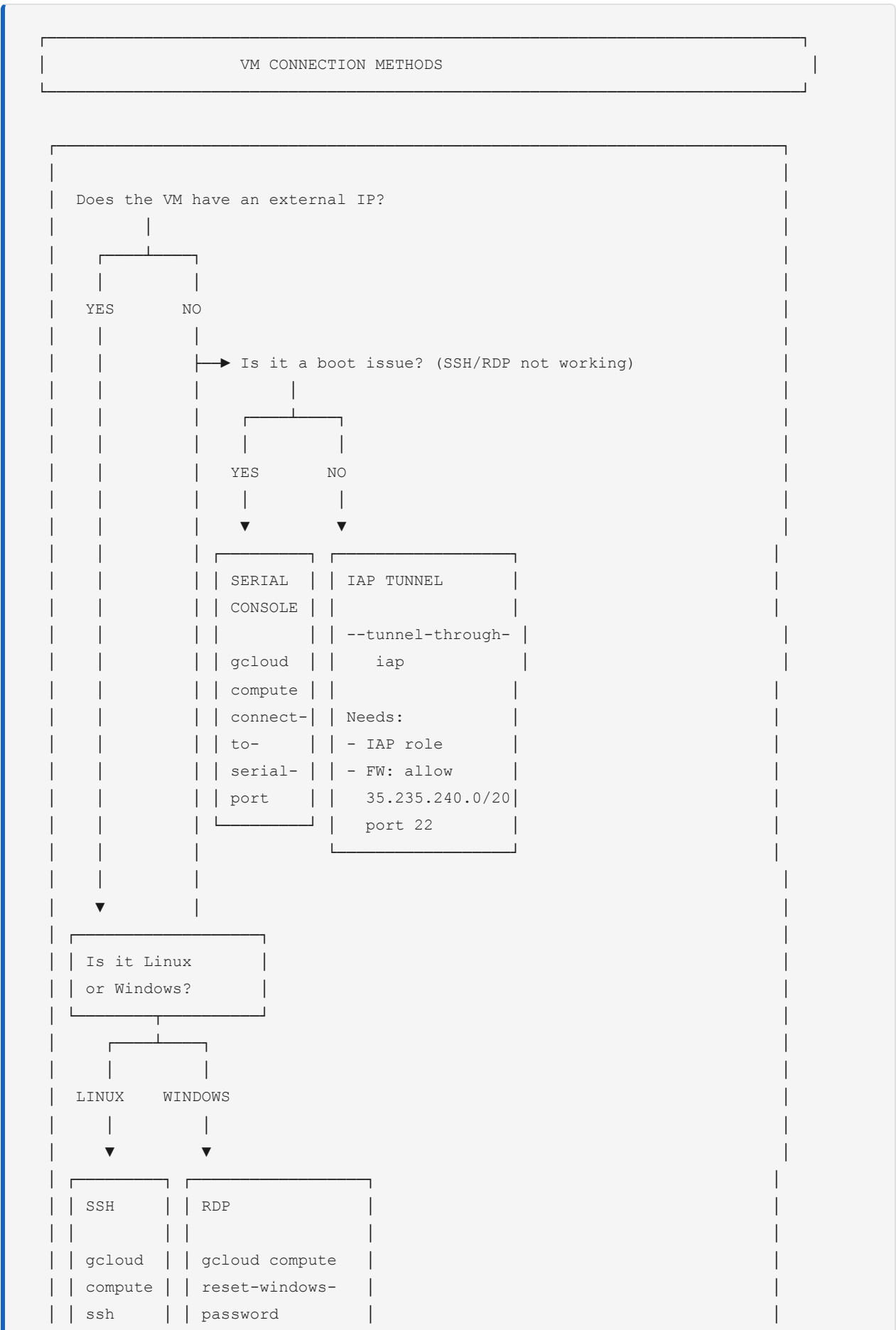
Private VMs (no external IPs)



CLOUD DNS ZONE TYPES:

PUBLIC	PRIVATE	FORWARDING	PEERING
Visible to the entire internet	Visible to specified VPCs only	Forward to on-premises DNS servers	Query another VPC's DNS
External services	Internal services	Hybrid cloud	Cross-project

11. VM Connection Methods

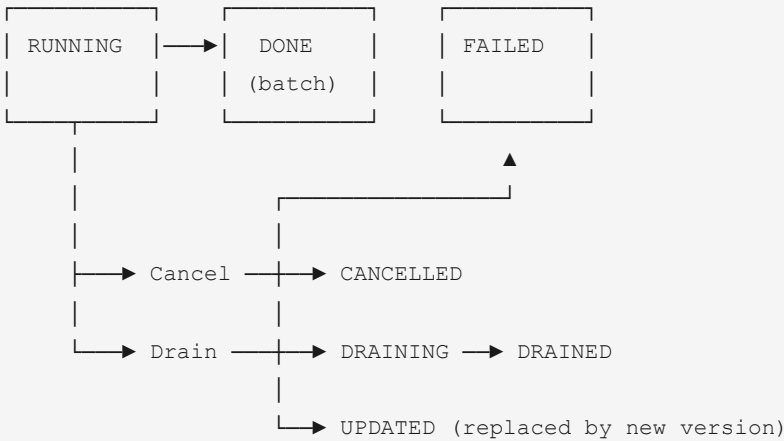




12. Dataflow and BigQuery Job Status

JOB STATUS MONITORING

DATAFLOW JOB LIFECYCLE:



Cancel vs Drain:

CANCEL:	Stops immediately. In-flight data may be lost.
DRAIN:	Processes remaining data, then stops gracefully.
	Use drain for streaming jobs to avoid data loss.

BIGQUERY JOB TYPES:

QUERY	LOAD	EXTRACT	COPY
SQL execution	Import data	Export data	Table to table

Cost estimation (before running):

```
bq query --dry_run 'SELECT * FROM ...'
```

Returns: Bytes to be scanned (no cost, no execution)

Use this to estimate cost = bytes scanned * \$5/TB

